

Bridge Design — Two Evidence Options

NAME _____ DATE _____ PERIOD _____

Constraints and Evidence Boundary

- 12-inch clear span between secured supports
- teacher-provided preparked material limit
- one marked load point
- same known staged loads and stop rule for every physical test
- equal fixed-data route when the physical test gate is not met

This classroom prototype does not validate a real bridge. Real work requires licensed analysis, codes, materials testing, geotechnical evidence, public review, and longer validation.

Bridge Systems

- **Beam:** deck/member spans supports; performance depends on geometry, material, connections, span, and loading.
- **Truss:** connected members form triangular units that can stabilize an idealized pin-jointed frame; joint/member behavior still matters.
- **Arch:** curved form transfers compression toward supports/abutments; support behavior matters.

Option A — Top View

Label 12-inch span, supports, members, joints, and marked load point.



Option A — Side View and Prediction

Expected load path and predicted weak point:

Option B — Top View

Label 12-inch span, supports, members, joints, and marked load point.



Option B — Side View and Prediction

Expected load path and predicted weak point:

Critique, Choice, and Career Role

Evidence-based strength of Option A:

Evidence-based strength of Option B:

Chosen option and reason:

Career/role responsible for the next step, evidence needed, and work product:
